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Scams, Spoofing: Govt's Concerns Over WhatsApp's Username Feature

Scams, spoofing: Govt's concerns over WhatsApp's username feature

Soumyarendra Barik
New Delhi, July 2

THE CENTRE has asked Meta-owned WhatsApp not to roll out its upcoming username feature over concerns of impersonation and fraud. In a notice issued to the company on Wednesday, the IT Ministry has given WhatsApp three days to furnish a detailed explanation of the feature.

This week, WhatsApp announced it would gradually allow users to select a username and hide their phone numbers when contacting another user for the first time. It is currently allowing users to "reserve" a username to avoid overlaps with others. It is an optional unique identifier that can be used to message or call, while keeping a user's phone number private. Messaging services like Telegram and Signal have similar features.

Though the company is billing the feature as one making WhatsApp conversations more private, some are concerned that bad actors could use it for impersonation. WhatsApp earlier clarified in response to the government's notice that it had built safeguards around the feature.

Username as identity

In its notice, the government is understood to have told WhatsApp that the feature may materially increase the incidence of online fraud, phishing and digital arrest scams. The government has also flagged that the feature may facilitate impersonation of public authorities and government agencies, by permitting the adoption of usernames closely resembling those of genuine people and institutions.

Senior government officials said a risk assessment of the new feature would be undertaken. "During our internal assessment and subsequent questioning of WhatsApp, if we do not think there are enough guardrails to protect against impersonation, the government may also look into whether the feature's rollout should be stopped altogether," an official said, requesting anonymity.

PRIVACY CONCERN

- WhatsApp says the feature, which will allow contact through usernames instead of phone numbers, secures privacy.

- But it could potentially enable bad actors to misrepresent their identities, opening another avenue for scams.

India is WhatsApp's biggest market, with over 850 million users. Some in the Indian tech community had also criticised the feature. Mohilvika's CEO Bipin Preet Singh said on X: "Not a good idea at all. Will lead to proliferation of fraud and impersonation. For example, I checked, most variations of my name already taken. Wonder what can it be used for."

WhatsApp's defence

A WhatsApp spokesperson said that to protect against impersonation, the company has held high-profile names of public figures, government entities and celebrities.

"Users still require a phone number to use WhatsApp and we've built multiple layers of defense against scams into usernames. Other users need to know the exact username to message you, we will limit how many new people an account can contact... and have systems to detect and remove activity showing common impersonation and abuse patterns," they added.

In an earlier blog post announcing the feature, WhatsApp said: "Usernames are our latest step to make WhatsApp even more private. There's no directory to browse and no suggestions." As an added safety feature and to allow users to control who can reach them on WhatsApp with their username, the platform has built an optional numeric key that others will need to know to message a user.

"When you message someone for the first time who doesn't have you saved in their contacts, we might share whether your phone number is registered in a country different from theirs to help them know who they're connecting with so they can stay safe and avoid unwanted contact," WhatsApp said.

The feature has not yet been rolled out. For creators, small businesses, and organisations that may want to maintain a consistent presence online, WhatsApp will allow an option to claim their existing Instagram or Facebook username on WhatsApp. The company said it takes impersonation-related abuse "seriously," and will take action against users who may pretend to be someone else.

Scams, Spoofing: Govt's Concerns Over WhatsApp's Username Feature

- **The New Feature**

- WhatsApp plans to let users be contacted using a unique username instead of their phone number, an optional identifier similar to features on Telegram and Signal

- **Government's Concern**

- The IT Ministry issued a notice asking WhatsApp not to roll out the feature over risks of impersonation, phishing and "digital arrest" scams, seeking an explanation within three days

- **Impersonation Risk**

- Usernames resembling genuine public officials or institutions could facilitate fraud; a government risk assessment of the feature is underway

- **WhatsApp's Safeguards**

- The company says it has reserved high-profile names, limits new contact requests, and still requires a phone number to use the app

- **Scale of the Issue**

- India is WhatsApp's largest market with over 850 million users; industry voices have flagged the risk of large-scale username impersonation

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Cyber Warfare:

- **Cyber Warfare refers to a virtual conflict initiated as a politically motivated attack on the enemy's computer and information systems waged via internet.**
- **The aim of such an intrusion usually is to disrupt the systems, destroy them partially or entirely.**
- **For instance, In 2010, Stuxnet, a malicious computer worm, was designed to attack industrial programmable logic controllers of the Iranian nuclear programme.**

Cyber Espionage:

- **Use of computer networks to gain illicit access to confidential information, typically held by a government or other organizations.**
- **For instance, in Operation Shady RAT, the hackers had access to seventy government and private agencies around the world**

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Cybercrime:

- **Cyber Stalking** : The use of the internet, email or other electronic communications to stalk a person is called as cyber stalking.
- **Denial of Service** : A denial-of-service attack is a technology-driven attack that occurs when an attacker prevents legitimate users from accessing specific computer systems and networks.
- **Malware** : It is a malicious software designed to perform an unwanted illegal act via the computer network. Some of the common malwares are – viruses, worms, trojans, hoax etc.
- **Phishing** : It is a cybercrime in which targets are lured by mails to provide sensitive information(personal information, bank details) by someone posing as a legitimate website.
- **Botnet** : Botnets are a number of internet-connected devices, with each one running its own bots. They are used to perform DoS attacks, steal data, send spam mails etc.

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Cyber Terrorism:

- **Politically motivated use of computers and IT to cause severe disruption or widespread fear in society is called as cyber terrorism.**
- **It includes activities such as websites spreading extremist propaganda, recruiting terrorists, promoting the propaganda of terrorists etc.**

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Cyber Crimes

Authorized Push Payment- Deceiving victims into voluntarily transferring funds to fraudulent accounts. Eg- “Digital Arrest” scams in 2024-25 – losses of over .

Ransomware-as-a-Service (RaaS)- Deploying malicious software that encrypts critical data, demanding payment for the key.

Cyber-Slavery – Trafficking individuals to foreign “fraud factories” to operate transnational scam centers. Eg- “Cyber Slavery” hubs in Cambodia and Myanmar

Man-in-the-Middle (MitM) Attacks- Intercepting and potentially altering communications between two parties without their knowledge.

Vikram-1, India's First-Ever Privately Developed Orbital Rocket, Gears Up for Launch

Vikram-1, India's first-ever privately developed orbital rocket, gears up for launch

The Hindu Bureau
BENGALURU

India's first privately developed orbital-class rocket is all set for its maiden launch.

On Thursday, private space launch company Skyroot Aerospace, based in Hyderabad, announced that the launch window for Vikram-1's first test flight, named Mission Aagaman, is between July 12 and August 4.

It said that Test Flight-1 is targeted for no earlier than July 12, subject to the completion of assembly and testing operations at the launch site at the Satish Dhawan Space Centre in



Vikram-1 sits on a launch pad at the Satish Dhawan Space Centre in Sriharikota.

Sriharikota, as well as weather, safety and range clearance. The launch window extends till August 4.

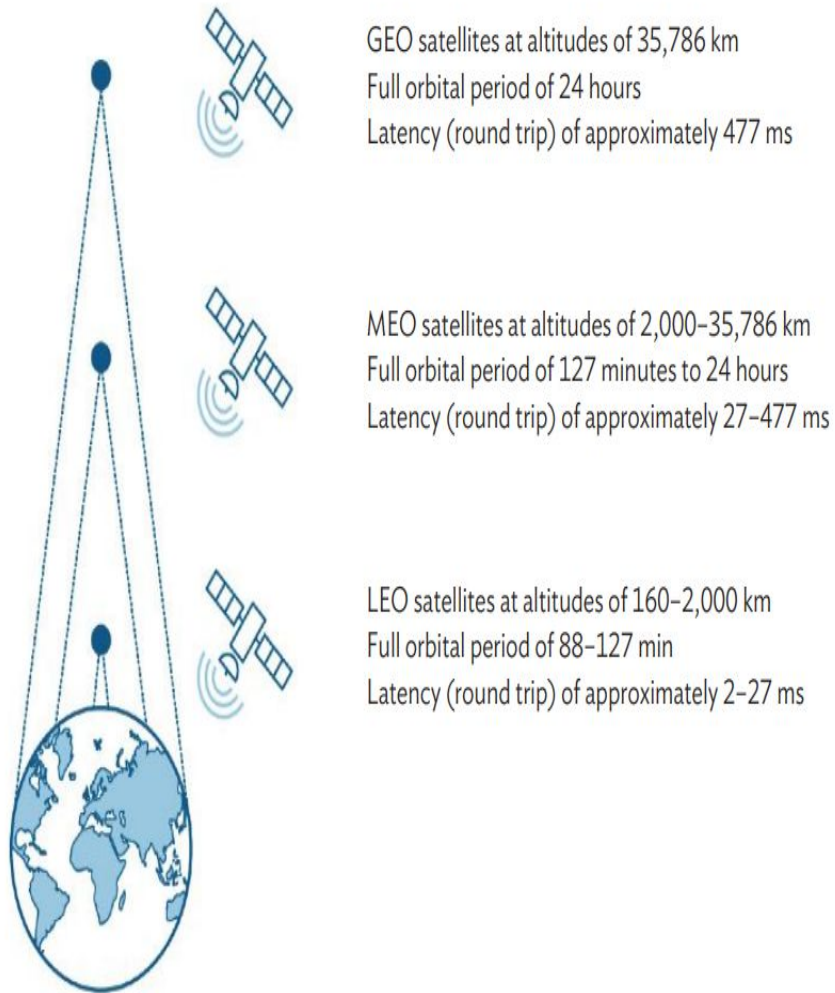
The company said that Mission Aagaman, meaning 'the arrival', marks its

second mission following the successful suborbital flight of Vikram-S, the first private rocket to reach space from Indian soil, on November 18, 2022.

It said that this would be a partially commercial flight, with the company planning to commence full commercial flights after one or two successful demonstrations to orbit.

All stages of the Skyroot's Vikram-1 have been successfully integrated and stacked at the launch pad. The mission will gather critical data across propulsion, stage separation, guidance, navigation, control and overall vehicle performance.

Vikram-1, India's First-Ever Privately Developed Orbital Rocket, Gears Up for Launch



● Types of Orbits

- HEO lies at and beyond 35,786 kilometers from Earth's surface. It is a sort of “sweet spot” in which satellite orbit matches the rotation of the Earth.
- MEO lies between 2,000 km to 35780 km from the surface of the Earth. Satellites in MEO have an orbital period between 2 to 24 hours. This is the orbit used by the Global Positioning System (GPS) satellites
- LEO has an altitude between 160 km to 1000 km above the Earth's surface. Satellites in this orbit take approximately 88 - 127 minutes to circle Earth. This orbit is commonly used for satellite imaging, Earth observation.

Vikram-1, India's First-Ever Privately Developed Orbital Rocket, Gears Up for Launch

- **The Rocket**

- Vikram-1, developed by Hyderabad-based Skyroot Aerospace, is India's first privately developed orbital-class rocket

- **Mission Aagaman**

- Test Flight-1's launch window is between July 12 and August 4, from the Satish Dhawan Space Centre, Sriharikota

- **Predecessor — Vikram-S**

- Skyroot's earlier suborbital flight, Vikram-S, was the first private rocket to reach space from Indian soil, launched on November 18, 2022

- **Mission Objectives**

- Will gather critical data on propulsion, stage separation, guidance, navigation, control and overall vehicle performance

- **Commercial Roadmap**

- Planned as a partially commercial flight; full commercial operations expected after one or two successful demonstrations to orbit

Vikram-1, India's First-Ever Privately Developed Orbital Rocket, Gears Up for Launch

- **About Vikram-1**

- India's first privately developed orbital-class launch vehicle.
- Developed by Skyroot Aerospace, Hyderabad.
- Height: About 24 metres (seven-storey tall).
- Configuration: Multi-stage launch vehicle.
- Payload Capacity: Up to 350 kg to Low Earth Orbit (LEO).
- Target Orbit: 450 km altitude with 60° orbital inclination.
- Built using an all-carbon composite structure for lightweight construction.
- Solid & Liquid Propulsion: Vikram-1 uses a mix of solid and liquid propulsion systems developed entirely by Skyroot. The first three stages utilize powerful solid rocket motors named after Dr. A.P.J. Abdul Kalam.

How Temples Deal with Donations

How temples deal with donations

India's biggest temples have similar processes to handle donations. What differs are the institutions governing these systems



DEEPTIMAN TIWARY

EVERY DAY thousands of devotees fold currency notes into temple *handis* (donation boxes) believing the offering has reached the deity.

What follows is a donation handling chain that is broadly similar across India's biggest temples. Offerings are removed by authorised personnel, shifted to counting centres, segregated into cash, coins and valuables, counted, and recorded before being deposited into designated bank accounts. The entire process is under CCTV surveillance.

The recent allegations of theft of donations at the Ram Jammabhoomi Temple in Ayodhya have brought this invisible chain into public view. But Ayodhya is far from the country's only temple handling massive offerings. Shrines such as the Sri Venkateswara temple in Tirumala, the Jagannath temple in Puri, Vaishno Devi temple in Katra, Siddhivinayak temple in Mumbai and Kashi Vishwanath in Varanasi receive hundreds of crores in donations every year, besides tonnes of gold, silver and jewellery.

Where temples differ is not in the journey of the donation but in the institutions governing the handling process. We examine the donation systems, oversight mechanisms and governance frameworks of five of India's biggest temples and compare them with those at the Ram Temple.

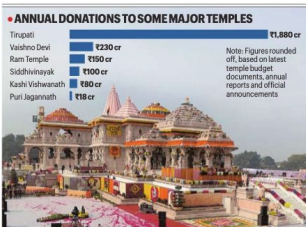
RAM TEMPLE: At the Ram Temple in Ayodhya, donations from about 35 *handis* are opened by authorised personnel, including trust officials and a State Bank of India (SBI) representative, and taken to a counting hall inside the pilgrim facilitation centre. There, counting staff outsourced by SBI assist staff count cash and jewellery under the supervision of retired banker Siddhant Srivastava, while overall responsibility for the donation process rests with trust member Anil Mishra. Once verified, the collections are deposited into the trust's SBI account.

TIRUMALA: Few religious institutions in the world process donations on the scale of the Balaji temple, and its famed 'Prankamani' system has evolved accordingly. Permanent Tirumala Tirupati Devasthanams (TTD) finance personnel, representatives of notified allied banks, and vetted volunteers — mostly serving or retired government and bank employees — work together to process donations under a CCTV network monitored by the temple's vigilance wing. People wearing pocketless clothes, are frisked before entering and leaving the counting hall. Cash movement is secured through armoured transport under escort, while access to different stages of the process is segregated among finance officials, vigilance staff, bank representatives and volunteers.

JAGANNATH: At Puri, *handis* are opened under the supervision of the temple administrator or an authorised gazetted officer, while a member of the Managing Committee serves as an independent witness.

Each *handi* is sealed before and after opening, entries are made in statutory forms, and CCTV monitors the handling of donations. Designated accounts maintain statutory registers for cash and valuables before bank deposits are made. The temple has also expanded official donation channels through payment gateways and Odisha's digital *handi* initiative.

VAISHNO DEVI: At Vaishno Devi, donation handling centres are the first corporate-style



The Ram Temple in Ayodhya on its consecration day in January 2024. (i)

• Systems in the states

TAMIL NADU: The Hindu Religious and Charitable Endowments Department, established under the Tamil Nadu Hindu Religious and Charitable Endowments Act, 1958, administers over 40,000 temples.

ANDHRA PRADESH: The Tirumala Tirupati Devasthanams (TTD) is an independent trust that manages the Sri Venkateswara Temple and several allied temples. The government appoints its Board of Trustees and Chief Executive Officer.

KERALA: The Travancore Devaswom Board, constituted in 1950, manages over a thousand temples, including the Sabarimala Temple.

UTTAR PRADESH: Following a major temple theft in January 1983, the UP Sri Kashi Vishwanath Temple Act was enacted by the then Congress government. A Board of Trustees was entrusted with its management, with a government-appointed CEO, JAMMU & KASHMIR. A state government-appointed body manages the Vaishno Devi temple.

management. Donation boxes are opened by committees comprising account officers, area managers, and security personnel rather than individual trustees or religious functionaries. Finance, security and operations are handled by specialised administrative departments under the Shrine Board. Given its mountainous location, the Board uses battery vehicles and, when required, helicopters, to move valuables.

SIDDHIVINAYAK: Here, independent oversight begins at the counting table itself. According to treasurer Pawan Tripathi, the main *handi* is opened every Thursday in the presence of an executive officer, a trustee, a bank representative, and an auditor under CCTV surveillance, ensuring multiple independent stakeholders witness every stage of the counting. Devotees also have access to official banking and digital donation channels.

KASHI VISHWANATH: At Kashi Vishwanath, the district administration is part of the accountability chain. The temple's 56 donation boxes are opened under the supervision of a Sub-Divisional Magistrate, while counting takes place in the presence of bank officials and a retired gazetted officer. Deposit receipts are generated for every transaction to create an audit trail. Jewellery is weighed by government-approved appraisers before being taken into custody.

How Ram Temple stands apart

Most of India's older major temples are administered under dedicated legislations enacted by state governments. TTD functions under the Andhra Pradesh Charitable and Hindu Religious Institutions and

Endowments Act, the Jagannath Temple under the Sri Jagannath Temple Act, the Shri Mata Vaishno Devi Shrine under the Jammu and Kashmir Shri Mata Vaishno Devi Shrine Act, the Siddhivinayak Temple under a Maharashtra law governing its trust, while the Kashi Vishwanath Temple is administered under the Uttar Pradesh Shri Kashi Vishwanath Temple Act.

These laws create the governing bodies, define the powers of administration, prescribe financial procedures, provide for government oversight and statutory audits, and in some cases lay down detailed rules for handling donations.

The Shri Ram Jammabhoomi Teerth Koharta functions through a trust, does not have a dedicated statute governing the administration of the temple. Day-to-day management, including appointments and financial administration, rests with the trust.

Another point of departure is the composition of the management. In several older temples, key financial processes involve executive officers, statutory administrators, government nominees, magistrates, auditors, or other public officials whose responsibilities are defined under law.

At the Ram Temple, many of the key officers associated with donation management — including General Secretary Champai Bui, trust member Anil Mishra, administrator Gopal Rao, and donation supervisor Subhash Srivastava — have long associations with the RSS or its affiliates. Responsibility for administration is therefore concentrated within the trust structure rather than distributed across a broader statutory administrative framework.

The distinction assumes significance because a private internal audit commissioned by the trust within months of its formation identified weaknesses in the administrative framework.

Unlike many temple boards, the Ram Temple trust is not subject to mandatory financial audits by the state or central governments. Questions relating to public audit and oversight of the trust's finances have also reached the courts. The trust has not publicly disclosed whether the recommendations contained in the 2020 audit were fully implemented.

When systems are tested

No major temple has been untouched by controversy. Tirumala has over the years tightened access controls, vigilance, and surveillance after instances of theft involving employees and volunteers. At Jagannath, the Ratna Bandhar dispute centred on the custody and inventory of temple valuables, eventually leading to court-directed scrutiny and fresh inventories.

At Kashi Vishwanath, efforts have increasingly focused on routing donations through official channels rather than direct offerings to priests, while Siddhivinayak has periodically faced scrutiny over governance and financial management.

The allegations at the Ram Temple are unique in their scale and the conclusions have been reached. But the experience of India's older temples suggests institutional safeguards are rarely built overnight.

The traditional models to manage temples, state's role



SHYAMLAL YADAV

THE RECENT allegations of embezzlement of offerings and donations at the Ram Jammabhoomi Temple in Ayodhya have brought its management into focus. The Shri Ram Jammabhoomi Teerth Koharta Trust was formed in 2020, when the Union Cabinet cleared the way for its establishment.

With reports of irregularities, Ram Temple Construction Committee Chairman Nripendra Mishra has advocated for appointing a CEO to manage trust affairs. On the other hand, the Vishwa Hindu Parishad, the main driving force behind the Ram Temple movement, has called for "liberating" temples across the country from government control.

Temples and their assets

There are no official figures for Hindu temples in India, but estimates put the number at around 10 lakh. Most of these are small village shrines or modest structures maintained by local communities, families or hereditary priests who take the offerings and donations. A significant number of large temples own substantial properties and receive large donations from devotees. Charitable trusts often manage temples. Significant donations in the form of cash offerings and goods in kind are mostly unaccounted for.

Three main traditions of temple management

Temples are managed through diverse systems, primarily under three traditional models, with varying degrees of government involvement.

1. FAMILY MANAGEMENT (PANDAS) AND PRIEST FAMILIES

The majority of temples are managed by specific families or hereditary priest lineages known as *pandas* or *purohits*. Offerings, donations and ritual responsibilities traditionally belong to them. When multiple families are involved, control often rotates.

A notable example is the Udupi Sri Krishna Mutt in Karnataka. It is administered by the *Adhva Mathu* or eight monasteries founded by the 18th-century saint Madhacharya. Each *matha* manages temple administration and rituals for two years, thus, the next cycle for a *matha* comes only after 16 years.

2. MAHANT SYSTEM

Here, a single spiritual head (*mahant*, *mathadish* or *piethadheeswar*) holds prime authority. The *mahant* exercises control over temple assets, properties, offerings, and administration and typically appoints or nominates his successor.

A prominent example is the Gorakhnath Math in Gorakhpur, UP, headed by Chief Minister Yogi Adityanath, who was appointed by the late Mahant Avaidyanath. Similar success-based systems operate in the Shankaracharya *mathas*.

3. ANADVA PANCHAYATI SYSTEM

Adhavas are autonomous organisations of *adhas*. These are often in the limelight during the Mahakumbh, where they are prioritised for the holy dip according to their relative status. *Adhavas* function as collective bodies, often with elected or consensus-based heads and also called "Panchayati Adhava". They appoint priests, oversee rituals, and control donations.

Government's role in managing temples

The British introduced the Religious Endowments Act in 1863, which handed over control of temples to committees set up under the Act. However, the government retained considerable influence through other legal provisions.

In 1925, the Madras Hindu Religious Endowments Act empowered provincial governments to legislate on endowments. It was gradually expanded to include oversight, even allowing takeover of the temple management. After Independence, the 1925 Act became the blueprint for various state laws.

Article 26 of the Constitution empowers the state to regulate or restrict "any economic, financial, political or other secular activity which may be associated with religious practice" and to enact laws for social welfare and reform.

This provision is the basis for state legislation governing temple endowments, appointing and Christians manage their places of worship and religious institutions through community boards or trusts.

Apart from government-appointed bodies playing a role in temple management, they also pay salaries to priests and ensure that offerings and donations are deposited into dedicated accounts. States also use part of the income from offerings at large temples for the administration and upkeep of those and smaller temples, and for non-temple welfare activities.

How Temples Deal with Donations

- **The Trigger**

- Allegations of embezzlement of offerings at the Ram Janmabhoomi Temple, Ayodhya, brought temple donation management into public focus

- **Three Traditional Models of Temple Management**

- Family management (pandas/priest families), the Mahant system (a spiritual head controlling the temple estate), and the Akhada (Panchayati) system of ascetic organisations

- **State Legislative Role**

- Article 25(2) of the Constitution empowers the State to regulate secular activities associated with religious practice; states have enacted specific endowment laws (Tamil Nadu 1959, Andhra Pradesh's TTD Act, Kerala's Devaswom Boards)

- **Varying Governance Structures**

- Large temples like Tirupati and Vaishno Devi operate under dedicated statutory boards with CCTV oversight and audits; the Ram Temple Trust is not subject to mandatory state or central financial audits

Can Space Oncology Revolutionise Cancer Treatment?

Can space oncology revolutionise cancer treatment?

Even as the burden of cancer is rising in India, space oncology is a rapidly emerging field. Investigating how microgravity and cosmic radiation impact cancer progression and treatment

K. Ganapathy

Cancer is a group of diseases where abnormal cells grow uncontrollably, invading nearby tissues and sometimes spreading to distant organs of the body. In India, an estimated 1.47 million new cases are likely to be diagnosed in 2026, which means 1 in 9 Indians reading this article are at a lifetime risk of developing cancer. Damage wreaked on the body apart, cancer is one of the most economically disruptive illnesses for families. Every year, ₹3,400 crores is spent on direct and indirect medical costs for cancer; this, however, excludes income loss, debt, asset depletion and caregiver burden. It is no wonder then, that with the burden of cancer rising in India and globally, that it is one of the most widely-studied diseases in the world. And it is this research that has led to the emergence of what could perhaps become a new frontier in cancer treatment: space oncology.

What is space oncology?
Space oncology is a rapidly emerging field, investigating how microgravity and cosmic radiation impact cancer progression and treatment. Space is a natural laboratory for studying cancer biology. Space-based environments are studied to accelerate tumour modelling and drug discovery. Space oncology has already produced useful, conceptual and technological spin-offs, ranging from 3D cell-culture systems to protein-crystal studies for cancer drug development. This is in addition to refined thinking about radiation risk, tissue response and biomarker discovery.

Why this is useful?
Cancer is biologically diverse. Understanding divergent mechanobiological responses helps identify signalling pathways fundamental to metastasis, tissue invasion and treatment resistance. Some cell changes that have been identified as a result of microgravity are cytoskeletal reorganisation, altered focal-adhesion signalling, changes in extracellular-matrix interactions and spontaneous formation of multicellular spheroids. Each is relevant to tumour spread and therapeutic response. When it comes to drugs, researchers have found that microgravity changes the cytoskeleton and spheroid formation of cells, which can help hasten drug discovery and also lead to reduced animal testing for drugs.

As microgravity also allows for more uniform protein crystals and lower-viscosity biologics, this could help develop more stable formulations.

Microgravity also changes the tumour-cell biology that drugs act on, and improves performance of delivery systems such as nanoparticles and 3D formulations. The effects of microgravity on cancer cells, cancer stem cells, and drug response therefore, could help future therapeutic strategies.

International Space Station (ISS) studies have covered real-space and

simulated microgravity experiments across breast, lung, thyroid, prostate, melanoma, glioblastoma and hematologic cancer models. Cancer cells studied in microgravity and exposed to unique radiation environment of space behave differently than in conventional laboratory systems. These differences reveal mechanisms that are otherwise difficult to understand.

Reviews of real-space and simulated microgravity experiments show effects on cell adhesion, migration, proliferation, gene expression and formation of multicellular spheroids.

On microgravity
Microgravity also changes how tiny drug packages (nanoparticles) form. These particles carry chemotherapy directly into the tumours and release it slowly. Microgravity reorganises cancer-cell shapes, membrane behaviour and gene expression. This alters how cells take up drugs. Remove or reduce gravity, and cells experience major changes in fluid behaviour, mechanical loading and cell-to-cell interaction.

For cancer cells, these changes are not trivial. In microgravity, protein crystals and complex biologics form more slowly and uniformly without gravitational sedimentation. Microgravity, therefore, helps produce drug crystals and biologic formulations that are harder to make on Earth.

Breast & GI cancer in outer space
Breast cancer cells show changes in gene expression, morphology, signal transduction and invasive behaviour under microgravity. In a microgravity

environment of space, breast cancer cells generally shift toward a less malignant and less aggressive phenotype. Under normal gravity, breast cancer cells attach strongly to their surrounding matrix using focal adhesions. In space, these adhesion points fail to mature, reducing the cells' ability to crawl, migrate, and metastasise. Key proteins that regulate cell cycle

checkpoints, specifically cyclin D1 and cyclin B1, are heavily downregulated, pausing the division process and stopping the rapid colony-forming ability of the tumour. Breast cancer spheroids grown in space are more susceptible to specialised therapies.

Gastrointestinal and colorectal cancer, however, react to microgravity by accelerating their disease trajectory and become more aggressive. Reduced expression of drug resistance genes, increased DNA/RNA damage markers, and reorganisation of the protein, F-actin, make gastric cancer cells more sensitive to the chemotherapy drug, doxorubicin.

Regulations and investments
In 2025, the United States' Food and Drug Administration (FDA) approved a subcutaneous form of an immunotherapy drug, pembrolizumab.

The U.S. space agency, NASA, had developed this through protein crystal growth research performed on the ISS, targeting the ADAM gene. These crystals were more uniform and better suited to supporting formulation work for this route of delivery.

Similarly, rebecicicab, became the first space-tested cancer drug to enter clinical trials. Following successful ISS-linked testing it received the FDA 'Investigational New Drug' status. Microgravity-grown tumour organoids helped demonstrate antitumor activity strong enough to support further regulatory progression.

These cases show that space-based research is now a legitimate pathway for drug development, not just a novelty. This year, United Kingdom regulators and the U.K. Space Agency publicly backed the development of a regulatory pathway in outer space.

The Medicine and Healthcare Products Regulatory Agency (MHRA), the Civil Aviation Authority (CAA), and the Regulatory Innovation Office have streamlined inter-agency bureaucracy. The specific 'dual-regulation' hurdles that previously made space-based

pharmacology legally risky for heavy corporate investment were addressed.

This has enabled startups to engineer highly stable, concentrated cancer therapies that patients can self-inject at home instead of hours-long IV infusions.

For companies, the costs of setting up a factory are removed; instead they may be able to deploy compact units that operate independently in orbit.

The Indian Space Research Organisation (ISRO), in spite of a modest \$13 billion space economy, is the third-largest space-tech power globally and fifth among major government space agencies, falling launch costs and new commercial platforms are making space oncology and space pharmaceutical manufacturing more realistic. With successful space startups in India the prices may fall further.

Space manufacturing is no longer about research. The picture is clear: the industry is moving from proof-of-concept to commercial production.

The microgravity pharmaceutical manufacturing market valued at \$1.6 billion in 2025 is projected to reach \$9.8 billion by 2034. Commercial space stations and nanomicrosatellites (CubeSats) are now essential platforms for experiments. Small, frequent launches are becoming viable for pharmaceutical payloads.

What the future holds
Over 700 peer-reviewed papers, 40 chapters and 12 monographs have already been published in the area of space oncology. At present, space oncology can be encapsulated by the well-known saying: 'A journey of a thousand miles begins with the first step.' There is no doubt that humans will exploit outer space in the perpetually quest to solve the cancer conundrum on planet Earth.

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When it comes to drugs, researchers have found that microgravity changes the cytoskeleton and spheroid formation of cells, which can help hasten drug discovery and also lead to reduced animal testing for drugs. As microgravity also allows for more uniform protein crystals and lower-viscosity biologics, this could help develop more stable formulations.



Latest advances: Cancer is biologically diverse. Understanding divergent mechanobiological responses helps identify signalling pathways fundamental to metastasis, tissue invasion and treatment resistance. (iStockphoto)

THE GIST

Space oncology is a rapidly emerging field, investigating how microgravity and cosmic radiation impact cancer progression and treatment. Space is a natural laboratory for studying cancer biology. Space-based environments are studied to accelerate tumour modelling and drug discovery.

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Can Space Oncology Revolutionise Cancer Treatment?

- **What is Space Oncology?**

- An emerging field studying how microgravity and cosmic radiation affect cancer progression and treatment, using space as a natural laboratory

- **Effects of Microgravity on Cancer Cells**

- Alters cytoskeletal structure, cell adhesion and gene expression; breast and GI cancers respond differently — some become less aggressive, others more so, under microgravity

- **Drug Development Benefits**

- Microgravity produces more uniform protein crystals and lower-viscosity biologics, aiding more stable drug formulations and reducing animal testing

- **Global Regulatory Momentum**

- The US FDA approved a subcutaneous immunotherapy drug developed via ISS-based protein crystal research; UK regulators are developing a dedicated regulatory pathway for space-based drug development

- **India's Position**

- ISRO, despite a modest \$13 billion space economy, is the third-largest space-tech power globally; falling launch costs are making space pharmaceutical research more accessible

A Shot at Life: Mandsaur's Model for HPV Vaccination

A shot at life, Mandsaur's model for HPV vaccination

The birth of Savita (name changed) in Mandsaur district, Madhya Pradesh, 14 years ago was a moment of celebration for her family. The Banchhada community she was born into – a denotified tribe in Madhya Pradesh traditionally associated with sex work – welcomes the birth of girls, who are often viewed as future breadwinners. Yet, when a team of human papillomavirus (HPV) vaccinators recently approached Savita's family, they were apprehensive. "Will she be able to work?" they asked candidly, worried about the vaccine's future impact.

Their hesitation was neither unexpected nor isolated. Although cervical cancer is the second most common cancer among Indian women, preventive health-care initiatives often grapple with low levels of awareness and social stigma. Cervical cancer vaccination faces a distinct set of challenges – low cultural sensitivity around sexual health and gender bias, coupled with vaccine hesitancy – making it a pressing public health concern.

In light of this, the Government of India launched a nationwide cervical cancer campaign on February 28, 2026, providing free HPV vaccinations to 1.15 crore girls aged 14-15 years. India bears a quarter of the global cervical cancer burden, reporting over 1.2 lakh new cases and 80,000 deaths annually. Since nearly 95% of cases are caused by high-risk HPV strains, vaccination offers a significant preventive breakthrough.

From data to coverage

To implement the programme effectively and inclusively, the Mandsaur district administration adopted a data-driven, decentralised and adaptive strategy. For exhaustive coverage, the most vulnerable and often overlooked populations were targeted first.

Girls from difficult-to-reach communities – Banchhadas, nomadic tribes, urban slums, and school dropouts – became the starting point. These "missed populations" are at greater risk of falling off the radar of government service delivery.

At the grassroots, the challenge is often not vaccine hesitancy but data invisibility. Leveraging



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This district in
Madhya Pradesh
has
demonstrated
effective
grassroots
strategies for
HPV vaccination
delivery

multiple government databases – Rashtriya Bal Swasthya Karyakram (RBSK), SAMAGRA MP (a citizen-centric social security platform by the Government of Madhya Pradesh) and Ladli Laxmi Yojana hyper-localised target lists were created. Fragmented records were transformed into actionable intelligence. Door-to-door surveys and tracking through SAMAGRA IDs ensured that eligible girls did not fall through statistical gaps. School and anganwadi enrolment gaps over the years were meticulously analysed to prepare village-level Master Line Lists. Rapid identification of 'low-coverage/high-resistance' areas enabled coordinated micro-planning. Mapping vulnerable groups geographically also enabled customising communication strategies, tailored to cultural sensitivities.

Reducing barriers through 'nudges'

Behavioural insights played a pivotal role in grassroots' saturation. The district relied on the "Nudge Approach" that helped design an environment where vaccination became the default choice. This helped in effectively breaking down grassroots barriers such as reluctance, inertia, social discomfort. Health-care workers informed families that their daughters were 'due for vaccination', rather than asking them to choose to vaccinate. Families who refused vaccinations received repeated counselling visits by health-care teams. Schools and local bodies arranged transportation to eliminate logistical barriers, simplifying access.

Misinformation hoaxes and myths around vaccine-induced infertility resulted in initial resistance. To counter rumours, the administration launched targeted awareness campaigns involving Gen-Z influencers and youth icons. National-level athletes, young doctors, students, religious leaders and media personalities voluntarily joined the effort to promote vaccination and dispel misconceptions.

Social norms and peer networks served as powerful "behavioural nudges". Local events publicly felicitated vaccinated families and recognised vaccinated girls as peer champions. Gram panchayat and ward-level data sharing sparked both collaboration and competition. Recognition was given to both top performers

and the most improved. "Digital nudges" and red-flag reminders for frontline workers enhanced monitoring and accountability.

At the cutting-edge implementation level, resistance often manifests as delay, doubt, and discomfort rather than outright refusal. Vaccinations were conducted exclusively under medical supervision at visible health-care facilities, normalising the practice of adolescent girls' vaccination. Experiences of women affected by cervical cancer were shared to create emotional resonance and encourage informed decision-making. Counselling sessions reduced stigma and replaced distrust with dialogue.

Strategy to impact

The HPV campaign was further integrated with on-going health-care programmes. Routine immunisation days, antenatal care clinics and the Pradhan Mantri Surakshit Matritva Abhiyan sessions were used to promote HPV awareness and conduct vaccination drives. When women accessed one service, they became more receptive to another, creating avenues for 'bundling' of health-care service delivery.

The results were significant. In less than 40 days, Mandsaur achieved 100% of its vaccination target – 493 vaccination sessions were conducted through 12 permanent and 27 temporary vaccination sites across the district. Girls who met the criteria, from 893 villages and 190 urban wards, were mobilised, moving from planning on paper to protecting the population.

The magnitude of India's health-care challenge must be met with the measure of its grassroots' actions. Policy design must bridge the yawning gap between intended outcomes and empirical ground realities. Behavioural and systemic "nudges" in implementation can leverage the human tendency to opt for preset choices. Dovetailing health-care data and grassroots innovations tailored to regional realities can bridge the last mile and deliver the final dose.

By moving from coverage to care, from data to impact, Mandsaur turned a formidable public health challenge into a collective mass movement. The district's inclusive vaccination campaign demonstrated how a simple shot in the arm can truly become a fair shot at life.

A Shot at Life: Mandsaur's Model for HPV Vaccination

- **The Challenge**

- Cervical cancer is the second most common cancer among Indian women; India bears a quarter of the global burden, with over 1.2 lakh new cases and 80,000 deaths annually

- **National Campaign**

- Government of India launched a nationwide cervical cancer campaign on February 28, 2026, providing free HPV vaccination to 1.15 crore girls aged 14-15

- **Mandsaur's Data-Driven Strategy**

- District administration merged multiple databases (RBSK, SAMAGRA MP, Ladli Laxmi Yojana) to create hyper-localised, village-level target lists and identify "low-coverage/high-resistance" areas

- **The 'Nudge Approach'**

- Behavioural nudges made vaccination the default choice; health workers framed vaccination as girls being "due", rather than seeking consent, to reduce hesitancy

- **Results**

- Mandsaur achieved 100% of its vaccination target in under 40 days through 493 sessions across 12 permanent and 27 temporary sites, covering 893 villages and 190 urban wards

A Hold on AI: Scientific Promise, But Not Unchecked

A hold on AI

AI holds scientific promise, but it should not unfold unchecked

The UN's Preliminary Report of the Independent International Scientific Panel on AI drives at a few fault lines in the rapid investment into and proliferation of AI technologies: the Global South-Global North divide, with the latter poised to take the lion's share of the benefits of the diffusion of advanced AI models in different industries; and the challenge poorer countries face in regulating models far more advanced than what their own AI ecosystems can develop. These divides force emerging countries to choose between capital-intensive undertakings to get a seat at the table, or to accept the hand dealt to them by the half-dozen companies whose decisions shape AI access and use. These divisions are all the more urgent when seen in the light of what makes a country a voice that matters in AI: abundant electricity, highly capitalised firms that employ scarce talent with high salaries, and a drive to plow through policy resistance with a competitive zeal to outdo last month's capabilities. While the report highlights AI technologies' transformative potential in scientific research, its development is far more rapid than that of social media – an industry that had time to act responsibly yet deformed public discourse and affected democracies, even as people were more or less able to tell what was going on in real time. It is unclear whether individual countries, even China or the U.S., have enough power to meaningfully arrest how AI develops.

AI firms acting irresponsibly have already caused harms that would not be tolerated in any other industry. AI models have ensnared teenagers and adults in parasocial fantasies that have sometimes turned fatal; they have flattened the world wide web, grievously injuring the news media's ability to deploy resources in their respective missions to inform the public; they have unleashed an epidemic of deepfakes, undermining trust in the written word and images alike; and they have deeply intertwined their promises of Artificial General Intelligence with global financial systems, with possible catastrophic economic consequences. The most important task for governments is how to hold these firms to account when needed, as taking a backseat on this conversation – even if industry leaders ignore the Global South's concerns – carries far greater costs. India has already experienced the cost of not asserting itself in the technology development of the decade: even as Anthropic's Mythos and Fable held out the tantalising promise of securing vast cyber systems, the U.S. pulled access, leaving the firms that had these models to ponder the risk of a different model succeeding at attacking their infrastructure. There is a limit to visions of AI as a geostrategic asset; much needs to be done to address its potential for broad damage.

A Hold on AI: Scientific Promise, But Not Unchecked

- **The UN Report**

- The UN's Preliminary Report of the Independent International Scientific Panel on AI highlights key global fault lines in AI development

- **The Global Divide**

- A Global South-Global North gap, with wealthier nations positioned to capture most benefits of advanced AI diffusion; poorer countries struggle to regulate models more advanced than their own ecosystems

- **What Confers AI Power**

- Abundant electricity, highly capitalised firms with scarce talent, and a competitive drive to outpace rivals define which countries have a meaningful voice in AI governance

- **Documented Harms**

- Parasocial AI relationships with sometimes fatal outcomes, disruption of the news media's revenue base, a deepfake epidemic, and entanglement of AI promises with global financial systems

- **The Governance Challenge**

- Even major powers may lack the ability to meaningfully control AI's pace of development; governments face the urgent task of holding AI firms accountable

Thank You